

Convergent Institutional Evolution: Organized Crime as Natural Experiment for the Extended Phenotype Theory of Law

Ignacio Adrián Lerer

Independent Researcher

Buenos Aires, Argentina

adrian@lerer.com.ar

ORCID 0009-0007-6378-9749

March 2026

Abstract

Why do criminal organizations separated by centuries, continents, and cultures independently develop the same institutional trajectory: genuine provision of dispute resolution and property protection, followed by parasitic capture and extraction?

This paper argues that the transcultural convergence of organized crime constitutes a natural experiment for Extended Phenotype Theory (EPT) as applied to legal institutions. Drawing on econometric studies of the Sicilian Mafia (Dimico et al. 2017; Buonanno et al. 2015; Bandiera 2003), comparative analyses of Russian violent entrepreneurship (Volkov 2002), Japanese Yakuza governance (Hill 2003; Milhaupt and West 2000), Hong Kong Triads (Wang 2023; Chu 2000), California prison gangs (Skarbek 2014), Latin American criminal governance (Lessing 2021; Arjona 2016), and London gang governance (Campana 2025), I identify seven independent cases exhibiting functional convergence.

The paper reinterprets the five transition mechanisms documented in the literature (time-horizon shortening, threat endogenization, progressive monopolization, active institutional deficit perpetuation, and state-criminal collusion) as predictions derivable from replicator dynamics within the EPT framework. It proposes that the honest-signal degradation that transforms protectors into parasites in criminal organizations operates through the same mechanism that degrades signal quality in formalized legal systems, connecting the Coalitional Intelligence Hypothesis (Egeland et al. 2026) to institutional rigidity as measured by the Constitutional Lock-in Index.

Three falsifiable predictions distinguish this framework from existing accounts. The analysis suggests that organized crime does not merely fill governance vacuums; it recapitulates, in observable time scales, the evolutionary dynamics that produced formal legal institutions over millennia.

Keywords: extended phenotype theory, evolutionary game theory, organized crime, institutional evolution, criminal governance, convergent evolution, replicator dynamics, honest signaling, constitutional lock-in, legal origins

JEL Codes: K00, K42, D74, P48, B52

1. Introduction

In 1872, Dr. Gaspare Galati inherited a citrus grove near Palermo and discovered that his estate warden, Benedetto Carollo, was systematically depressing revenues. When Galati dismissed Carollo, the local mafia boss Antonino Giammona orchestrated the assassination of two replacement wardens and a campaign of ritualistic threats that forced Galati to abandon the property (Dickie 2004). The mechanism was precise: the mafia manufactured the violent threat that made its own protection services indispensable. What had begun as a genuine security arrangement between landowner and armed guard had transformed into a closed parasitic loop.

Twelve thousand kilometers away and 120 years later, the Primeiro Comando da Capital (PCC) was founded in São Paulo's prison system as a human rights defense organization following the 1992 Carandiru massacre. The PCC developed an elaborate quasi-judicial system, the *juízo*, in which parties presented cases before PCC arbitrators who issued binding decisions under a written code of conduct (the *proceder*). By establishing a monopoly on authorized violence within prisons and subsequently exporting this governance model to favelas, the PCC achieved a measurable reduction in homicide rates that econometric analysis attributes causally to the organization's governance function (Biderman et al. 2014; Lessing and Willis 2019).

The puzzle is not that criminal organizations provide governance. Gambetta (1993), Varese (2001, 2013), Tilly (1985), and Olson (1993) have documented this extensively. The puzzle is convergence. Organizations with no historical, cultural, or organizational connection, operating across five continents and four centuries, independently develop the same institutional trajectory: emergence in governance vacuums, provision of genuine dispute resolution and property protection, progressive capture by the provider, and transition to parasitic extraction. Sicilian *gabellotti*, Russian enforcement partnerships, Japanese *tekiya*, Hong Kong Triad headmen, Californian prison gang constitutions, Latin American narco-governance systems, and London governance gangs all follow this pattern (Skaperdas 2001; Milhaupt and West 2000; Wang 2023; Skarbek 2014; Campana 2025).

Existing theoretical frameworks capture important dimensions of this phenomenon. Olson's (1993) stationary bandit model explains why territorial monopolists develop encompassing interests. Tilly's (1985) state-mafia analogy reveals the structural equivalence between state formation and

organized crime. Gambetta's (1993) protection-industry thesis identifies the core commodity. Bandiera's (2003) property-rights model formalizes the demand-side conditions. Varese's (2013) time-horizon analysis specifies the transition mechanism. Skarbek's (2014) extralegal governance framework generalizes beyond mafia contexts.

What none of these frameworks provides is a unified selectionist account of why convergence occurs, why the protection-to-parasitism trajectory is invariant, and why the same degradation pattern appears in both criminal organizations and formal legal systems. This paper proposes that Extended Phenotype Theory (EPT), combined with Evolutionary Game Theory (EGT) and the Coalitional Intelligence Hypothesis (CIH), supplies that account.

The argument proceeds as follows. Section 2 develops the theoretical framework, positioning organized crime as a natural experiment for EPT. Section 3 presents the convergence evidence across seven independent cases. Section 4 reinterprets the five transition mechanisms documented in the literature as predictions derivable from replicator dynamics. Section 5 identifies the honest-signal degradation paradox that connects criminal organizations to formal legal institutions. Section 6 derives three falsifiable predictions that distinguish this framework from existing accounts. Section 7 discusses implications and limitations.

2. Theoretical Framework: EPT, EGT, and the Convergent Evolution of Institutions

2.1 Extended Phenotype Theory Applied to Legal Institutions

The Extended Phenotype Theory of law (Lerer 2025a, 2026a) treats legal norms as cultural replicators, memes in Dawkins's (1982) original technical sense, that are subject to variation, differential replication, and selection. Legal institutions, on this account, are extended phenotypes: structures built by normative memes to ensure their own transmission, analogous to the beaver dam that Dawkins (1982) analyzed as an extended phenotype of beaver genes. Just as the dam modifies the environment in ways that enhance the replicative success of the genes that built it, a constitutional court modifies the institutional environment in ways that enhance the replicative success of the normative memes it enforces.

This framework generates a specific prediction about institutional convergence. In biological evolution, convergent evolution occurs when unrelated organisms independently develop similar traits in response to analogous selection pressures (e.g., the independent evolution of eyes in vertebrates and cephalopods). If legal institutions are extended phenotypes of normative replicators, then analogous selection pressures should produce analogous institutional phenotypes regardless of cultural substrate. The same functional problem (coordination under conditions of

absent state enforcement) should generate the same institutional solution (private protection backed by credible violence) through independent evolutionary paths.

2.2 The Coalitional Intelligence Hypothesis as Microfoundation

Egeland, Kennair, and Kleppetø (2026) propose in *Evolution and Human Behavior* that human intelligence evolved as an honest signal of coalitional value. Individuals who provided superior cognitive-computational services to their group, including conflict resolution, collective coordination, and resource access facilitation, received prestige and deference in return. The key mechanism is honest signaling: coalitional value can be assessed with reasonable accuracy even from brief interactions (Borkenau et al. 2004; Driebe et al. 2021), which prevents cheap exploitation of the prestige system by low-value individuals.

I have argued elsewhere (Lerer 2026b) that the CIH provides the microfoundation for EPT: the cognitive-computational services described by Egeland et al. are, functionally, proto-legal services. The formalization of these services into transmissible, enforceable rules is the mechanism by which coalitional value exchange scaled beyond small groups. The Roman jurisconsult, the hunter-gatherer mediator, and the contemporary judge are institutional variants of the same evolutionary pattern.

For the present analysis, the CIH contributes a critical mechanism: honest-signal degradation. In small groups where feedback is direct and reputational consequences are immediate, the coalitional value system rewards genuine problem-solving capacity. When institutions formalize and scale, three factors degrade signal quality: (a) evaluation shifts to proxies (credentials, publications, appointments) that can diverge from actual problem-solving capacity; (b) recognition decouples temporally from service provision; and (c) the dominant memplex can capture the prestige-allocation mechanism, rewarding replication fidelity over innovative competence (Lerer 2026b). This degradation mechanism, I argue, operates identically in criminal organizations and in formal legal systems.

2.3 Evolutionary Game Theory and the Stationary Bandit Equilibrium

Olson's (1993) stationary bandit model can be reformulated in EGT terms as an evolutionarily stable strategy (ESS) analysis. Consider a population of coercive organizations competing for territorial control. Two strategies are available: roving banditry (extract maximum short-term rents and move on) and stationary governance (limit extraction to sustainable rates and invest in territorial order). The payoff structure depends on time horizon: organizations with long time horizons obtain higher lifetime payoffs from governance, while those with short time horizons obtain higher payoffs from roving predation.

The stationary governance strategy is an ESS when the following conditions are met: (a) territorial control is secure enough that the discount rate for future revenues is low; (b) the cost of providing

governance is less than the marginal increase in extractable surplus that governance generates; and (c) competitors adopting roving strategies can be excluded from the territory. When these conditions hold, a population of governance providers cannot be invaded by a mutant roving predator. This explains why criminal organizations that achieve territorial monopoly converge on governance provision: the stationary bandit equilibrium is an ESS under conditions of secure territorial control.

The transition from governance to parasitism can then be modeled as a perturbation that disrupts the ESS conditions. Police pressure shortens the effective time horizon (condition a). Competition from rival organizations increases the cost of territorial control (condition c). Internal organizational change (leadership succession, fragmentation) alters the discount rate. When any of these perturbations is sufficiently large, the governance ESS is destabilized and the organization transitions toward extractive strategies.

2.4 Why Convergence Is Structurally Necessary

The EPT framework predicts convergence through a mechanism that existing frameworks identify empirically but do not derive theoretically. The argument has four steps.

First, the production of enforcement services depends on the monopolistic control of violence within a territory (Lane 1958; Skaperdas 2001). Violence production exhibits economies of scale, creating natural monopoly tendencies regardless of organizational origin.

Second, any actor that achieves a monopoly on violence within a territory faces identical revenue-maximization problems. The optimal strategy (limiting extraction, investing in order, protecting the productive base) is determined by the payoff structure, not by the actor's cultural identity, ideological commitments, or organizational history.

Third, effective governance requires credible commitment mechanisms. Criminal organizations, like states, develop costly signals (initiation rituals, codes of silence, mutual hostage structures) that substitute for formal institutions (Gambetta 2009; Campana and Varese 2013). The same information problem (how to credibly commit to cooperation without formal enforcement) generates the same signaling solutions across unrelated organizational contexts.

Fourth, evolutionary selection among competing criminal organizations favors those that develop governance functions, because governance providers attract more clients, generate more sustainable revenue, and reduce interdiction risk relative to purely predatory competitors (Martinez-Vaquero et al. 2019). This is selection at the organizational level: governance-providing organizations outcompete extractive ones in territorial environments where long-term control is feasible.

Together, these four mechanisms constitute a structural theory of institutional convergence: any actor that successfully monopolizes violence within a territory, over a sufficiently long time

horizon, and in a context where state enforcement of property rights is incomplete, faces an incentive structure that generates the same core functions, protection, adjudication, extraction, and political management, regardless of origin or organizational form.

3. The Convergence Evidence: Seven Independent Cases

The empirical basis for the convergence hypothesis rests on seven cases that are independent in the strong sense: no historical, organizational, or cultural transmission connects them. Each emerged from a distinct governance vacuum, developed analogous institutional functions, and underwent a structurally similar transition toward parasitic extraction.

3.1 The Sicilian Mafia (Post-1860)

The Sicilian case is the most extensively documented, with three distinct evidentiary layers: historical-ethnographic (Franchetti 1877; Blok 1974; Lupo 1993; Sabetti 2002; Catanzaro 1992), economic-theoretical (Gambetta 1993; Varese 2013), and econometric (Dimico et al. 2017; Buonanno et al. 2015; Bandiera 2003).

The governance vacuum was created by the abolition of Sicilian feudalism in 1812 and the Italian unification of 1860-1861, which eliminated the feudal order as guarantor of property rights without replacing it with effective state institutions. The number of landowners increased from approximately 2,000 to 20,000 in a few decades, creating a mass of new property holders without legitimate protection mechanisms (Dimico et al. 2017). Leopoldo Franchetti, whose 1876-1877 investigation constitutes the first systematic analysis of the organization, identified the mafia as an *industria della violenza*: an industry of violence that filled the enforcement vacuum left by state failure (Dickie 2004).

The genuine service phase is documented with precision. The gabelotti system, in which intermediary administrators leased large estates from absentee landlords and subleased parcels to peasants, generated a demand for enforcement services that the state could not supply. Calogero Vizzini's career illustrates the pattern: beginning as a grain escort organizer at the request of a local priest, Vizzini expanded into general dispute resolution, eventually mediating between landowners, cooperatives, and the state during the 1950 agrarian reform (Sabetti 2002; Varese 2013). The *componenda*, a negotiated agreement for the restitution of stolen goods, functioned as a genuine arbitration service (Lupo 1993).

The parasitic transition operated through what Franchetti identified as the endogenization of threat: the same capacity directed against external aggressors could be turned against the very clients the organization claimed to protect. The case of Dr. Gaspare Galati (1872-1875) documents this mechanism with archival precision: when Galati dismissed a mafia-connected warden who was depressing revenues, the local boss orchestrated assassinations of replacement wardens until Galati

abandoned the property (Dickie 2004). The Antonio Ortoleva network in Mistretta (1920s) exemplified the closed parasitic loop: cattle thefts were secretly orchestrated by the armed guards connected to the mediation network that then charged fees for recovering the stolen livestock (Lupo 1993).

The econometric evidence confirms the demand-driven hypothesis. Dimico et al. (2017) use the exogenous shock of international citrus demand as an instrumental variable, showing that mafia presence at the municipal level is strongly associated with soil suitability for citrus cultivation, even controlling for other variables. Buonanno et al. (2015) document the same mechanism for sulfur mining: mafia emerged as a cartel of protection providers in areas where valuable resources created high demand for protection and state provision was low. Bandiera (2003) formalizes the collective-action problem: each landowner who purchased protection created negative externalities for others, generating a prisoner's dilemma structure that made the organization's persistence individually rational but collectively harmful.

3.2 Russian Organized Crime (Post-1991)

The collapse of the Soviet Union created the most rapid and complete governance vacuum in the modern documentary record. Volkov (2002) documents that state arbitration courts processed cases at three to four times the statutory delay, with enforcement rates estimated at only 32% for 1997. The number of organized crime groups grew from 952 in 1991 to 6,743 in 1996, and by 1997 criminal groups controlled or owned an estimated 40,000 enterprises, including more than 500 banks. Volkov introduces the concept of *silovoe partnerstvo* (enforcement partnership): a business function derived from the skilled use of real or potential force to maintain conditions of commercial activity, including security, contract enforcement, dispute resolution, and transaction insurance.

The genuine service dimension is documented through the emergence of *ponyatiinye soglasheniya* (notional agreements): a body of customary commercial law that businessmen actively incorporated into their contracts because the criminal enforcement system was faster, more flexible, and above all enforceable (Volkov 2002). The transition followed a three-stage model: from extraction (*poluchat'*), to control of the business (*kontrolirovat'*), to equity partnership (*byt' v доле*). The retreat of criminal governance came not primarily through law enforcement but through institutional reconstruction: Putin's 2000 tax reform and the expansion of state arbitration courts after 2002 reduced business informality and displaced criminal enforcement.

3.3 The Japanese Yakuza (Tokugawa Period to Present)

The yakuza case is analytically valuable because it refutes the state-failure hypothesis in its simple form. Japan is a wealthy democracy with strong formal institutions, yet it sustained extensive organized crime for centuries. Milhaupt and West (2000) demonstrate that yakuza presence

correlates negatively with civil litigation rates, bankruptcies, reported crimes, and outstanding bank credit across Japanese prefectures. The interpretation is that the yakuza filled *specific* enforcement gaps deliberately created by institutional design: a state-controlled number of attorneys (far too few for demand), extraordinarily slow and costly civil litigation, and police reluctance to intervene in private disputes.

Hill (2003) documents the yakuza's conflict mediation role through the concept of *minbo*, violent intervention in civil affairs: debt collection, bankruptcy management, real estate disputes, and traffic settlements where gang affiliation itself signaled coercive capacity sufficient to make outcomes stick. The *jiage* operations (land assembly for developers through coercive pressure on small-plot owners) represent a paradigmatic case of private ordering: developers hired yakuza to resolve holdout problems that formal property law handled too slowly, with commissions typically around 3% of inflated land value.

The honest signal dimension is documented in disaster relief operations. Following both the 1995 Kobe earthquake and the 2011 Tohoku disaster, yakuza organizations delivered water, food, blankets, and sanitary goods more quickly than government agencies, demonstrating retained logistics capacity and territorial embeddedness that served public-facing governance functions even in crisis conditions (Reuters 2011). Yakuza membership peaked at 184,000 in 1963 and declined following the 1992 anti-yakuza legislation, confirming the prediction that improved formal enforcement displaces informal governance.

3.4 Hong Kong Triads (Post-1842)

Wang (2023) proposes the concept of the criminal urban polity: a non-state power that supplies what the state declines to provide, including protection, social welfare, and basic living necessities. In colonial Hong Kong after 1842, the British government focused on commercial construction and largely ignored the livelihood of the Chinese population. By 1886, Triad membership reached 15,000-20,000 in a population of approximately 150,000. Colonial records document that residents preferred to submit their problems and disputes to their elected headmen, most of whom were triad members, rather than the established authorities (Wang 2023).

The historical origins of Triads in mainland China confirm the EPT prediction. The Tiandihui (Heaven and Earth Society) was founded in Fujian province around 1761-1762 as a mutual-aid and self-protection association of lower-class people (Ownby 1995). Average landholdings in Fujian had shrunk from 5.0 mu per person in 1571 to 0.93 mu by 1812, far below subsistence level, creating demand for collective protection that the Qing government did not supply (Wang and Kwok 2023). The Kongsis of West Borneo (late eighteenth to late nineteenth century) represent the purest case of Chinese migrant self-governance: sovereign federations with taxation, adjudication, commercial regulation, and external relations, organized through the same associational structures that in other settings produced Triad organizations.

3.5 California Prison Gangs (Post-1957)

Skarbek's (2011, 2014) analysis provides the most formally rigorous test of the demand-driven governance hypothesis, because prisons constitute a controlled institutional environment. Before the 1950s, California prisons were governed by the convict code: informal norms sustained by reputation in small, stable populations. Between 1944 and 1969, the inmate population grew from 5,710 to 27,535, exceeding the capacity of informal norms. Prison gangs emerged to provide the institutional infrastructure, territory, membership boundaries, enforcement mechanisms, dispute resolution procedures, that informal norms could not sustain in large, anonymous populations.

The Mexican Mafia's governance extends beyond prison walls. Skarbek (2011) documents that the organization extorts street-level drug dealers in Los Angeles because dealers anticipate future incarceration, giving the gang a credible enforcement threat over street transactions. This creates an externalized governance function that produces order in the urban narcotics trade. La Nuestra Familia's written constitution, with its separation of powers, judicial procedures, and prohibition of internal predation, represents a constitutional order within a total institution.

3.6 Latin American Criminal Governance

The Latin American evidence introduces a refinement to the governance-vacuum hypothesis. Uribe et al. (2026) document that criminal governance is more prevalent where the state is more present but perceived as corrupt or partial, not simply in vacuums of state absence. Between 13% and 14% of survey respondents across 18 Latin American countries, representing 77 to 101 million people, report living in areas where criminal groups provide order or reduce crime.

The PCC case documents the full lifecycle with exceptional clarity: founded as a human rights organization (1992), development of a quasi-judicial system with written codes and binding arbitration, export of governance from prisons to favelas, and measurable reduction in violent crime attributable to the organization's monopoly (Biderman et al. 2014). La Familia Michoacana offers the most systematically documented social contract: redistribution of taxes to poor families, construction of schools and drainage systems, and vigilante justice against Los Zetas, followed by the successor organization's (Caballeros Templarios) complete abandonment of welfare commitments and shift to massive predation, triggering the 2013 autodefensas uprising (Herrera 2023).

The Argentine case (Dewey 2015; Flom 2019) introduces a structural variant: criminal governance mediated by the state rather than substituting for it. The police and politicians sell protection to illegal markets in exchange for resources and political control. The state itself behaves as a mafia, selling protection and promoting demand in the underworld of illegal markets (Dewey 2015). This variant confirms that the convergence mechanism operates at the functional level, not at the organizational level: the same enforcement-for-extraction exchange occurs whether the provider is a criminal organization, a state agent, or a hybrid.

3.7 London Governance Gangs (2025)

Campana (2025) applies the criminal governance framework to gangs in a wealthy welfare state, documenting governance functions (dispute mediation, agreement enforcement, public space regulation) in London neighborhoods with poor-quality public services but sufficient residential stability for gangs to accumulate relational capital. This case confirms that functional convergence does not require state collapse: governance deficits in specific domains within otherwise strong states are sufficient to generate the same institutional response.

4. Five Transition Mechanisms as Predictions of Replicator Dynamics

The comparative literature identifies five distinct mechanisms through which criminal organizations transition from governance to parasitism. I reinterpret each as a prediction derivable from EPT and replicator dynamics, rather than as an ad hoc empirical generalization.

4.1 Time-Horizon Shortening

Varese (2013) and Olson (1993) identify the protector's time horizon as the pivotal variable. A protector with a long time horizon maximizes future rents through quality service; one with a short horizon maximizes immediate extraction. Police pressure, inter-group competition, and political instability all shorten effective time horizons.

EPT reinterpretation: In replicator dynamics, the time horizon determines whether governance or extraction is the ESS. When the discount rate rises above a threshold determined by the ratio of governance costs to extraction gains, the governance equilibrium destabilizes. The memes encoding governance norms (dispute resolution procedures, codes of conduct, investment in territorial order) lose replicative advantage relative to extraction memes (immediate rent-seeking, violence escalation, territorial abandonment). The transition is a phase change in the fitness landscape, not a strategic decision by a rational actor.

4.2 Threat Endogenization

Franchetti (1876) documented that Sicilian mafiosi were in a position to create their own demand. Skaperdas (2001) formalizes the mechanism: when a protection provider achieves monopoly, the same capacity that neutralizes external threats can fabricate them. The Galati case (Dickie 2004), the Ortoleva cattle-theft network (Lupo 1993), and the Camorra eco-mafia (Saviano 2006) all document this mechanism with archival precision.

EPT reinterpretation: Threat endogenization is a form of niche construction by parasitic memes. The extended phenotype (the protection apparatus) modifies the selective environment in ways

that increase the replicative success of the parasitic memes at the expense of the host population. This is structurally identical to the mechanism by which brood parasites (e.g., cuckoos) manipulate host behavior: the parasite constructs an environment in which the host's evolved responses (protection-seeking behavior) serve the parasite's replicative interests rather than the host's. The mafia that manufactures the threat it sells protection against is constructing a niche in which its normative memes (pay for protection, resolve disputes through us, do not cooperate with the state) have maximum replicative fitness.

4.3 Progressive Monopolization

Skaperdas (2001) demonstrates that competition in protection markets takes the form of violent turf wars rather than price competition, creating an inherent tendency toward monopolization. Once a protection monopoly is established, the competitive constraint on extraction is removed.

EPT reinterpretation: Monopolization is the institutional analogue of competitive exclusion in ecology (Gause's principle). When two normative systems (two criminal organizations, or a criminal organization and the state) compete for the same enforcement niche, the one with lower costs or greater coercive capacity excludes the other. The resulting monopoly eliminates the selection pressure that maintained governance quality, allowing parasitic drift: the progressive accumulation of extractive norms in the absence of competitive selection against them. This is the same mechanism that the Constitutional Lock-in Index measures in formal legal systems: when institutional actors face no external competition, internal status dynamics drive the system toward rigidity rather than adaptation (Lerer 2026a).

4.4 Active Institutional Deficit Perpetuation

Paoli (2003) documents that Sicilian cosche obstructed agricultural cooperatives and reform projects that would have reduced demand for their services. Trejo and Ley (2020) document that Mexican cartels actively capture or weaken municipal governments to maintain demand for private governance.

EPT reinterpretation: This mechanism is extended phenotype maintenance under EPT. Just as genes build and maintain the structures that ensure their continued replication, governance memes build and maintain the institutional conditions that ensure their continued dominance. When a criminal organization actively frustrates state reforms, it is maintaining the selective environment in which its normative memes have maximum replicative advantage. The mechanism is not intentional in the folk-psychological sense; it is selectionist: organizations that happen to maintain their institutional niches outcompete those that allow niche erosion. Dennett's (1995) concept of competence without comprehension applies: the system behaves as if it were intentionally maintaining the governance vacuum, but the mechanism is differential survival, not strategic planning.

4.5 State-Criminal Collusion and Capture

In every documented case, a phase of collusion between organized crime and state agents is observed: Triads and colonial police in Hong Kong, cartels and the PRI in Mexico, Russian criminals and KGB/FSB operatives, yakuza and local political elites, Sicilian mafia and Italian state functionaries.

EPT reinterpretation: Collusion is memetic hybridization. When criminal governance memes and state governance memes coexist in the same institutional environment, selection can favor hybrid strategies that combine elements of both. The Colombian Convivir program (Grajales 2017), which provided legal cover for paramilitary governance, and the Argentine clandestine order (Dewey 2015), in which state agents sell protection to illegal markets, are examples of stable hybrid equilibria. The hybrid persists because it offers higher fitness than either pure strategy: criminal organizations gain legal cover and reduced interdiction risk; state agents gain economic resources and political control. When the hybrid equilibrium is disrupted (by anti-corruption reform, political change, or international pressure), the system transitions to one of two attractors: either formal state governance displaces criminal governance, or criminal organizations intensify predation to compensate for lost state protection.

5. The Honest Signal Degradation Paradox

The Coalitional Intelligence Hypothesis (Egeland et al. 2026) proposes that human intelligence evolved as an honest signal of coalitional value in small-group environments where feedback was direct and reputational consequences were immediate. The EPT framework identifies a paradox: the very process of institutionalization that scales coalitional value exchange beyond small groups simultaneously degrades the honesty of the signals on which the system depends.

In early-stage criminal organizations, the honest-signal system operates as designed. Calogero Vizzini gained initial prestige through genuine service, organizing armed escorts for grain transport at the request of a local priest (Sabetti 2002). The PCC's founders gained legitimacy through genuine defense of prisoners' rights after the Carandiru massacre (Lessing and Willis 2019). The Bakassi Boys in Nigeria achieved rapid legitimacy through demonstrable effectiveness against armed robbery (Agbu 2004). In each case, prestige followed demonstrable problem-solving capacity, exactly as the CIH predicts.

As organizations formalize and scale, three degradation mechanisms intervene. First, evaluation shifts from direct observation of outcomes to observation of proxies: rank within the organization, reputation transmitted through networks, endorsement by established leaders. Second, temporal decoupling: the prestige earned during the genuine-service phase persists long after the organization has transitioned to extraction. The Sicilian mafia's social legitimacy in the mid-twentieth century depended substantially on the memory of genuine services provided in the post-

unification period, decades or generations earlier. Third, memeplex capture: the organization's internal norms and rituals come to reward loyalty and conformity rather than genuine problem-solving capacity. The omerta code, initiation rituals, and hierarchical structures that initially served as honest signals of commitment become mechanisms for excluding innovation and maintaining the dominance of incumbent memes.

The critical theoretical point is that this degradation mechanism is not specific to criminal organizations. It operates identically in formal legal systems. When law faculties evaluate conformity with dominant doctrine rather than capacity to solve coordination problems, when judicial appointment systems reward publications in specialized journals over demonstrated resolution skills, when bar associations maintain entry barriers that protect incumbent practitioners rather than improve service quality, the same honest-signal degradation is occurring. The Constitutional Lock-in Index (Lerer 2026a) measures the institutional manifestation of this degradation: Argentina's CLI of approximately 0.89 reflects a system where institutional actors compete for status within the existing architecture rather than against the external problems the system was designed to solve. This is runaway social selection applied to institutions, precisely the mechanism Alexander (1990) described at the biological level and Egeland et al. (2026) applied to intelligence evolution.

The implication is that organized crime and formal legal systems are not analytically distinct phenomena that happen to share superficial similarities. They are manifestations of the same underlying evolutionary dynamic at different points on the institutionalization spectrum. The protection-to-parasitism trajectory documented in criminal organizations is the accelerated, observable version of the formalization-to-rigidity trajectory documented in constitutional systems. Both are driven by the same mechanism: the progressive degradation of honest signals of coalitional value as institutions scale and formalize.

6. Falsifiable Predictions

The EPT framework generates three predictions that are not derivable from existing theoretical accounts and that can, in principle, be tested against available or obtainable data.

6.1 Prediction 1: Convergence Intensity as a Function of Governance Deficit Specificity

Existing frameworks predict that criminal governance emerges where the state is weak. EPT predicts something more precise: the degree of functional convergence between independent criminal organizations should be positively correlated with the specificity of the governance deficit they fill. Organizations filling identical enforcement gaps (e.g., contract enforcement in

commercial transactions) should exhibit higher functional convergence than organizations filling different types of gaps (e.g., contract enforcement versus territorial security).

Operationalization: Code the primary enforcement function of each documented criminal organization across the seven cases (dispute resolution, contract enforcement, property protection, personal security, market regulation). Measure functional convergence using organizational-feature similarity indices (e.g., Jaccard similarity across a defined set of institutional features: written codes, hierarchical structure, initiation rituals, dispute resolution mechanisms, taxation systems, territorial boundaries). Test whether organizations filling the same enforcement function exhibit higher similarity than those filling different functions, controlling for geographic proximity and temporal period.

Falsification criterion: If organizations filling identical enforcement functions do not exhibit higher functional similarity than those filling different functions, the convergence-by-analogous-selection-pressure hypothesis is falsified, and alternative explanations (cultural diffusion, organizational mimicry, coincidence) gain support.

6.2 Prediction 2: Time-to-Parasitism as a Function of Monopoly Speed

The replicator dynamics model predicts that the speed of transition from governance to parasitism should be positively correlated with the speed at which the organization achieves enforcement monopoly within its territory. Rapid monopolization removes competitive selection pressure quickly, allowing parasitic drift to begin earlier.

Operationalization: For each documented case, estimate (a) the time from organizational founding to territorial monopoly and (b) the time from territorial monopoly to documented parasitic transition. Plot the relationship. The PCC's rapid monopolization in São Paulo (approximately 5-7 years from founding to territorial dominance) should be associated with faster parasitic transition than the Sicilian Mafia's gradual consolidation (approximately 40-60 years from post-unification emergence to documented extractive dominance). The Stoppagghieri of Monreale, which transitioned from civic protection to criminal predation within a few years of founding (Dickie 2004), represent a rapid-monopolization case that should confirm the prediction.

Falsification criterion: If no positive correlation is observed between monopoly speed and parasitic transition speed across cases, the competitive-selection-pressure hypothesis requires revision.

6.3 Prediction 3: Signal Degradation Parallel Between Criminal and Formal Institutions

The honest-signal degradation mechanism predicts a structural parallel: institutional domains where prestige allocation has decoupled most severely from problem-solving capacity in formal

legal systems should be the same domains where criminal governance is most persistent. This follows because both phenomena are driven by the same mechanism (signal degradation under reduced competitive pressure).

Operationalization: Identify legal domains where (a) the CLI is highest (greatest institutional rigidity) and (b) criminal governance is most persistent. The prediction is that these domains overlap. In Argentina, labor law exhibits the highest measured CLI (≈ 0.89) and is also the domain where informal enforcement mechanisms (union-mediated dispute resolution outside formal channels, tolerance of selective non-enforcement) are most entrenched. In Japan, civil dispute resolution exhibits both the highest barriers to formal access (fewest attorneys per capita, slowest court proceedings) and the most persistent yakuza involvement (Milhaupt and West 2000). The prediction would be falsified if high-CLI domains and persistent-criminal-governance domains do not overlap systematically across jurisdictions.

7. Discussion

7.1 Relationship to Existing Literature

This paper does not claim to supersede the contributions of Gambetta, Varese, Tilly, Olson, Skaperdas, Skarbek, Lessing, or Arias. Each of these scholars identified mechanisms that are incorporated into the EPT framework. What the EPT framework adds is a unified selectionist account that explains why these mechanisms converge across independent cases, why the protection-to-parasitism trajectory is invariant, and why the same degradation pattern appears in both criminal organizations and formal legal systems.

The relationship to Tilly (1985) deserves particular attention. Tilly's argument that state formation and organized crime are structurally equivalent is confirmed and extended by the present analysis. EPT adds a direction of explanation: states and criminal organizations are not merely analogous; they are instances of the same evolutionary process operating at different scales and under different selection pressures. The Sicilian gabellotto and the Tokugawa tekiya are to the modern state what proto-eyes are to the camera eye of vertebrates: earlier stages of the same adaptive trajectory.

The relationship to Lessing (2021) is complementary. Lessing's duopoly framework (criminal organization and state as co-producers of governance) describes the equilibrium structure. EPT provides the dynamics: how the duopoly equilibrium is reached, how it is maintained, and how it is destabilized. The finding that criminal governance is positively correlated with state presence (Uribe et al. 2026; Blattman et al. 2024) is consistent with the EPT prediction that memetic competition intensifies in environments where both formal and informal normative systems operate, just as biological competition intensifies in ecologically productive environments.

7.2 Limitations

Several limitations constrain the present analysis. First, the EPT framework is applied retrospectively to documented cases rather than prospectively to predict novel outcomes. The falsifiable predictions in Section 6 are designed to address this limitation, but they have not yet been tested. Second, the quantitative indices used (CLI, IHR) are theoretically motivated estimations, not fully validated empirical measurements. Third, the seven cases, while independent, are not a representative sample of all criminal organizations; they were selected because they are the best-documented, introducing a survivorship bias toward organizations that achieved sufficient prominence to attract scholarly attention. Fourth, the biological analogies (convergent evolution, niche construction, honest signaling) are intended as structural homologies between selectionist processes, not as claims about shared biological mechanisms. The extent to which cultural and biological evolution are governed by the same formal dynamics remains a contested question in evolutionary theory (Henrich 2016; Mesoudi 2011).

7.3 Implications for Legal Theory and Governance Policy

For legal theory, the analysis suggests that the traditional boundary between legitimate legal institutions and criminal organizations is analytically less sharp than jurisprudential tradition assumes. If both are manifestations of the same evolutionary dynamic (coalitional value exchange formalized through normative replicators), then the relevant distinction is not legitimacy versus illegitimacy but the position on the institutionalization spectrum: from informal proto-norms (hunter-gatherer dispute resolution), through semi-formal enforcement (organized crime governance), to formal institutionalization (state law). The diagnostic question for any legal system is not whether it is legitimate in some a priori sense, but whether its prestige-allocation mechanisms still track genuine problem-solving capacity or have been captured by incumbent memes.

For governance policy, the analysis suggests that interventions targeting criminal organizations are unlikely to succeed if they address only the organization while leaving the governance vacuum intact. The invariance of the protection-to-parasitism trajectory across independent cases implies that eliminating one criminal governance provider without improving formal enforcement will generate demand for a replacement. The most effective interventions are those that improve formal enforcement in the specific domains where criminal governance operates, as the Japanese case confirms: anti-yakuza legislation combined with expansion of formal dispute resolution reduced yakuza membership from 184,000 in 1963 to current levels, while pure repression in Mexico (the Calderón militarization of 2006-2012) fragmented cartels without reducing criminal governance.

8. Conclusion

Organized crime does not merely fill governance vacuums. It recapitulates, in observable time scales, the evolutionary dynamics that produced formal legal institutions over millennia. The seven independent cases analyzed in this paper document a convergence that is functional, not cultural:

the same selection pressures (governance deficit, demand for coordination, supply of violence specialists) produce the same institutional solutions (private protection backed by credible violence, formalized dispute resolution, territorial taxation, progressive monopolization) regardless of cultural substrate.

Extended Phenotype Theory provides the unifying account. Legal norms, whether enforced by states or by criminal organizations, are cultural replicators that build institutional structures to ensure their own transmission. The protection-to-parasitism trajectory is the lifecycle of these replicators under conditions of reduced competitive selection: when monopolization removes the pressure to maintain service quality, parasitic memes accumulate and capture the institutional apparatus. The honest-signal degradation that transforms protectors into parasites operates through the same mechanism in criminal organizations and in formal legal systems, connecting the Coalitional Intelligence Hypothesis to institutional rigidity as measured by the Constitutional Lock-in Index.

The practical implication is a diagnostic framework. When a legal reform fails, the relevant question is not whether it was technically sound, but what normative replicators it is competing against. When criminal governance persists despite state intervention, the relevant question is not whether enforcement is sufficient, but whether formal institutions provide the same coordination services that criminal governance supplies. And when institutional rigidity prevents adaptation, the relevant question is not who benefits from rigidity, but whether the prestige-allocation system still tracks genuine coalitional value or has been captured by incumbent memes defending their own replicative success.

The mafioso as proto-jurist is not a metaphor. It is a description of evolutionary convergence operating in real time.

AI Disclosure. *AI tools were used as research support throughout this work, including locating sources, extracting information from the academic literature, and reviewing drafts for consistency. The thesis, theoretical framework, arguments, and conclusions are the original contribution of the author as an independent researcher.*

Epistemological Positioning. *This paper is situated within a Lakatosian research program (Lakatos 1978) whose hard core consists of three propositions: (a) legal norms are cultural replicators subject to evolutionary selection (Dawkins 1982; Dennett 1995); (b) legal institutions are extended phenotypes that these replicators construct to ensure their own transmission (Lerer 2025a, 2026a); and (c) the dynamics of institutional change can be modeled using evolutionary game theory, specifically replicator dynamics and evolutionarily stable strategy analysis (Maynard Smith 1982). Popper (1959) provides the demarcation criterion: all predictions are formulated as falsifiable hypotheses. Kuhn (1962) identifies the object of study: organized crime as an anomaly that existing jurisprudential paradigms have not integrated into their explanatory apparatus. Hull (1988) and Campbell (1974) supply the meta-theoretical warrant for applying selectionist reasoning to cultural and institutional phenomena.*

References

- Agbu, O. (2004). "Ethnic Militias and the Threat to Democracy in Post-Transition Nigeria." Nordiska Afrikainstitutet Research Report No. 127.
- Alexander, R. D. (1990). *How Did Humans Evolve? Reflections on the Uniquely Unique Species*. Museum of Zoology, University of Michigan.
- Arjona, A. (2016). *Rebelocracy: Social Order in the Colombian Civil War*. Cambridge University Press.
- Bandiera, O. (2003). "Land Reform, the Market for Protection, and the Origins of the Sicilian Mafia." *Journal of Law, Economics, and Organization*, 19(1), 218-244.
- Biderman, C., De Lima, R. S., De Mello, J. M. P., and Schneider, A. (2014). "Pax Monopolista and Crime: The Case of the Emergence of the Primeiro Comando da Capital in São Paulo." CAF Working Paper.
- Blattman, C., Duncan, G., Lessing, B., and Tobón, S. (2024). "Gang Rule: Understanding and Countering Criminal Governance." *Review of Economic Studies*, forthcoming.
- Blok, A. (1974). *The Mafia of a Sicilian Village, 1860-1960: A Study of Violent Peasant Entrepreneurs*. Harper and Row.
- Borkenau, P., Mauer, N., Riemann, R., Spinath, F. M., and Angleitner, A. (2004). "Thin Slices of Behavior as Cues of Personality and Intelligence." *Journal of Personality and Social Psychology*, 86(4), 599-614.
- Buonanno, P., Durante, R., Prarolo, G., and Vanin, P. (2015). "Poor Institutions, Rich Mines: Resource Curse in the Origins of the Sicilian Mafia." *Economic Journal*, 125(586), F175-F202.
- Buss, D. M., et al. (2020). "Human Status Criteria: Sex Differences and Similarities across 14 Nations." *Journal of Personality and Social Psychology*, 119(5), 979-998.
- Campana, P. (2025). "Criminal Governance in a Large European City: The Case of Gangs in London." *Criminology and Criminal Justice*.
- Campana, P. and Varese, F. (2013). "Cooperation in Criminal Organizations: Kinship and Violence as Credible Commitments." *Rationality and Society*, 25(3), 263-289.
- Campbell, D. T. (1974). "Evolutionary Epistemology." In P. A. Schilpp (Ed.), *The Philosophy of Karl Popper*. Open Court.
- Catanzaro, R. (1992). *Men of Respect: A Social History of the Sicilian Mafia*. (Trans. R. Rosenthal). Free Press.
- Chu, Y. K. (2000). *The Triads as Business*. Routledge.
- Dawkins, R. (1982). *The Extended Phenotype: The Long Reach of the Gene*. Oxford University Press.
- Dennett, D. C. (1995). *Darwin's Dangerous Idea: Evolution and the Meanings of Life*. Simon and Schuster.
- Dewey, M. (2015). *El Orden Clandestino: Política, fuerzas de seguridad y mercados ilegales en la Argentina*. Katz Editores.
- Dickie, J. (2004). *Cosa Nostra: A History of the Sicilian Mafia*. Palgrave Macmillan.

- Dimico, A., Isopi, A., and Olsson, O. (2017). "Origins of the Sicilian Mafia: The Market for Lemons." *Journal of Economic History*, 77(4), 1083-1115.
- Driebe, J. C., Stern, J., Penke, L., and Gerlach, T. M. (2021). "Intelligence Can Be Detected but Is Not Found Attractive in Videos and Live Interactions." *Evolution and Human Behavior*, 42(6), 507-516.
- Egeland, H. T., Kennair, L. E. O., and Kleppesø, T. H. (2026). "Coalitional Psychology and the Evolution of Intelligence." *Evolution and Human Behavior*, 47, 106839.
- Flom, H. (2019). "State Regulation of Organized Crime: Politicians, Police, and Drug Trafficking in Argentina." *Latin American Politics and Society*, 61(3), 104-130.
- Franchetti, L. (1877). *Condizioni politiche ed amministrative della Sicilia*. Vallecchi (1974 reprint).
- Gambetta, D. (1993). *The Sicilian Mafia: The Business of Private Protection*. Harvard University Press.
- Gambetta, D. (2009). *Codes of the Underworld: How Criminals Communicate*. Princeton University Press.
- Garfield, Z. H. and Hagen, E. H. (2024). "Universal and Variable Leadership Dimensions across Human Societies." *Evolution and Human Behavior*, 45(5), 106602.
- Grajales, J. (2017). "Private Security and Paramilitarism in Colombia." *Journal of Politics in Latin America*, 9(3), 27-48.
- Henrich, J. (2016). *The Secret of Our Success: How Culture Is Driving Human Evolution*. Princeton University Press.
- Herrera, J. S. (2023). "Rejecting the Social Contract: Criminal Governance, Agrarian Inequalities and the Autodefensa Movement in Michoacán, Mexico." *Journal of Latin American Studies*, 55(1), 1-26.
- Hill, P. (2003). "Heisei Yakuza: Burst Bubble and Botaiho." *Social Science Japan Journal*, 6(1), 1-18.
- Hull, D. L. (1988). *Science as a Process: An Evolutionary Account of the Social and Conceptual Development of Science*. University of Chicago Press.
- Kuhn, T. S. (1962). *The Structure of Scientific Revolutions*. University of Chicago Press.
- Lakatos, I. (1978). *The Methodology of Scientific Research Programmes*. Cambridge University Press.
- Lane, F. C. (1958). "The Economic Consequences of Organized Violence." *Journal of Economic History*, 18(4), 401-417.
- Lerer, I. A. (2025a). "Before the Command Was Spoken: Pre-Linguistic Proto-Norms as Extended Phenotypes of Behavioral Memes." Zenodo. SSRN 5695662.
- Lerer, I. A. (2025b). "The Altruism Paradox in Law: Extended Phenotype Theory and the Evolution of Cooperative Legal Orders." Zenodo. SSRN 5478466.
- Lerer, I. A. (2026a). "Law as a Primary Adaptive Platform: Generalized Darwinism, Extended Phenotype Theory, and the Methodological Foundations of Evolutionary Jurisprudence." Zenodo. DOI: 10.5281/zenodo.18870552.

- Lerer, I. A. (2026b). "Strengthening the Foundations of Law as Extended Phenotype: Coalitional Intelligence and the Evolutionary Origins of Legal Institutions." Zenodo. DOI: 10.5281/zenodo.19104281.
- Lerer, I. A. (2026c). "Punctuated Institutional Equilibria: Stasis as Evolutionarily Stable Strategy, Basin Depth, and the Temporal Dynamics of Constitutional Lock-in." Zenodo. DOI: 10.5281/zenodo.19077323.
- Lessing, B. (2021). "Conceptualizing Criminal Governance." *Perspectives on Politics*, 19(3), 854-873.
- Lessing, B. and Willis, G. D. (2019). "Legitimacy in Criminal Governance: Managing a Drug Empire from Behind Bars." *American Political Science Review*, 113(2), 584-606.
- Lupo, S. (1993). *Storia della mafia*. Donzelli Editore.
- Martinez-Vaquero, L. A., Dolci, V., and Trianni, V. (2019). "Evolutionary Dynamics of Organised Crime and Terrorist Networks." *PLoS ONE*, 14(7), e0218553.
- Maynard Smith, J. (1982). *Evolution and the Theory of Games*. Cambridge University Press.
- Mesoudi, A. (2011). *Cultural Evolution: How Darwinian Theory Can Explain Human Culture and Synthesize the Social Sciences*. University of Chicago Press.
- Milhaupt, C. and West, M. (2000). "The Dark Side of Private Ordering: An Institutional and Empirical Analysis of Organized Crime." *Columbia Law Review*, 100(8), 2160-2238.
- Olson, M. (1993). "Dictatorship, Democracy, and Development." *American Political Science Review*, 87(3), 567-576.
- Ownby, D. (1995). "The Heaven and Earth Society as Popular Religion." *Journal of Asian Studies*, 54(4), 1023-1046.
- Paoli, L. (2003). *Mafia Brotherhoods: Organized Crime, Italian Style*. Oxford University Press.
- Popper, K. (1959). *The Logic of Scientific Discovery*. Routledge.
- Sabetti, F. (2002). *Village Politics and the Mafia in Sicily*. McGill-Queen's University Press.
- Saviano, R. (2006). *Gomorra: viaggio nell'impero economico e nel sogno di dominio della camorra*. Mondadori.
- Skaperdas, S. (2001). "The Political Economy of Organized Crime: Providing Protection When the State Does Not." *Economics of Governance*, 2(3), 173-202.
- Skarbek, D. (2011). "Governance and Prison Gangs." *American Political Science Review*, 105(4), 702-716.
- Skarbek, D. (2014). *The Social Order of the Underworld: How Prison Gangs Govern the American Penal System*. Oxford University Press.
- Tilly, C. (1985). "War Making and State Making as Organized Crime." In P. Evans, D. Rueschemeyer, and T. Skocpol (Eds.), *Bringing the State Back In*. Cambridge University Press.
- Trejo, G. and Ley, S. (2020). *Votes, Drugs, and Violence: The Political Logic of Criminal Wars in Mexico*. Cambridge University Press.

- Uribe, A. D., et al. (2026). "Criminal Governance in Latin America: Prevalence and Correlates." Perspectives on Politics.
- Varese, F. (2001). The Russian Mafia: Private Protection in a New Market Economy. Oxford University Press.
- Varese, F. (2013). "Protection and Extortion." In L. Paoli (Ed.), The Oxford Handbook of Organized Crime. Oxford University Press.
- Volkov, V. (2002). Violent Entrepreneurs: The Use of Force in the Making of Russian Capitalism. Cornell University Press.
- Wang, P. (2023). "Hong Kong Triads: The Historical and Political Evolution of Urban Criminal Polity." Urban History, 50(4).